

ARCHIT SHUKLA

Boulder, CO | +1 303 931 8281 | Archit.Shukla@colorado.edu | Portfolio

EDUCATION

- University Of Colorado Boulder** • Boulder, CO Aug 2024 - May 2026
Masters of Science • Computer Science • GPA: 3.91/4
Relevant Coursework: Generative AI for Software Development (LLMs, RAG, Agents), Machine Learning, Natural Language Processing, Computer Vision, Big Data Architecture (Distributed ML, Dataflow, Fault Tolerance), Design & Analysis of Algorithms
- SRM Institute of Science & Technology** • Chennai, India Aug 2020 - May 2024
Bachelor of Technology • Computer Science & Engineering • GPA: 3.81/4

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Go, SQL, JavaScript
- **Reinforcement Learning & Deep Learning:** PyTorch, TensorFlow, PPO, Transformer Architectures, Fine-Tuning, Transfer Learning, LoRA/QLoRA, Hyperparameter Tuning
- **GenAI & Agentic Systems:** RAG, Graph-Augmented Retrieval, Vector Databases (Qdrant), Embedding Models, LangChain, Dify, Hugging Face Transformers, vLLM, Ollama, Amazon Bedrock, Agent-Based Architectures
- **ML Systems & Serving:** Inference Pipelines, Model Deployment, Batch & Real-Time Serving, Experiment Tracking, GPU Acceleration (CUDA), Performance Optimization
- **Backend & Cloud:** Spring Boot, REST APIs, DynamoDB, PostgreSQL, MongoDB, Docker, CI/CD, Git, Linux, AWS (EC2, S3, IAM, Lambda, Bedrock), GCP

WORK EXPERIENCE

- Peleg Lab, BioFrontiers Institute** Aug 2025 – Jan 2026
Graduate Researcher—Distributed ML Systems Boulder, CO
- Increased model evaluation throughput by **35%** by building **distributed training and inference pipelines** in Python (PyTorch) on GCP with GPU acceleration and optimized data loading.
 - Fine-tuned **Cellpose segmentation models** using transfer learning and LoRA, improving inference stability and domain generalization through targeted preprocessing and loss tuning.
 - Accelerated experimentation cycles by **40%** by deploying **containerized, CUDA-enabled GPU training environments** with Docker, CI/CD debugging, and experiment tracking.
- CodeClause** June 2024 – July 2024
Software Engineering Intern—ML-Adjacent Backend Systems Bangalore, India
- Improved backend throughput by **20%** by optimizing **Flask-based REST APIs and SQL queries** supporting data-driven services and ML-powered analytics.
 - Reduced release failures by **25%** by automating **build, test, and deployment workflows** using Jenkins CI/CD and GitHub for model-backed services.
 - Enhanced service reliability through structured **unit testing, logging, and monitoring**, stabilizing downstream analytics and inference pipelines.
- Adflix Media** Apr 2023 – Sep 2023
Software Engineer Intern—Data Infrastructure Pune, India
- Processed **5M+ records daily** by scaling **SQL-driven ETL pipelines** via Dockerized services to support analytics and modeling workflows.
 - Reduced issue detection time by **30%** by implementing **Grafana dashboards** for latency and throughput monitoring.
 - Decreased recovery time by **25%** through structured logging & cloud monitoring, improving pipeline transparency & resilience.
- University of Colorado Boulder** Jan 2026 – Present
Teaching Assistant—Lab Systems Boulder, CO
- Led **3 EBIO 1240 lab sections** (60+ students), owning end-to-end execution and real-time issue resolution-improving outcomes.
 - Applied **statistical modeling & data visualization** (R; regression, hypothesis testing) to support data-driven instruction.

PROJECTS

- Regime-Switching Multi-Agent RL | Portfolio Optimization RL + Financial Modeling** Jan 2026 – Present
- Architected a **regime-aware multi-agent RL framework** in PyTorch, training regime-specific PPO agents with dynamic policy switching to handle market non-stationarity.
 - Increased risk-adjusted performance by integrating **drawdown penalties and Sharpe-based reward shaping**, surpassing single-policy RL and mean-variance baselines in backtesting.
 - Built a regime detection module using volatility clustering and rolling statistical features to enable adaptive asset allocation across market conditions.
- RepoGraph V2 – Distributed Debugging Platform** *Static Analysis + Graph-Based ML Systems*[Paper] Aug 2025 – Dec 2025
- Engineered a **distributed Python system** modeling execution order, shared state, and dependency graphs to localize and repair flaky tests across **100K+ LOC** repositories.
 - Constructed **graph-based dependency closures** exposed via REST APIs, enabling coordinated analysis, patch generation, and verification during large-scale evaluations.
 - Cut inter-service data movement by **70–85%** through graph-scoped retrieval, improving throughput and stability.
- Real-Time Anomaly Detection Service** *Streaming Data + ML Inference* [Demo] July 2025 – Aug 2025
- Implemented a **cloud-native streaming pipeline** processing **2K+ records per minute** with bounded latency for inference.
 - Trained a **PyTorch-based autoencoder anomaly detection model**, improving detection accuracy by **30%** over statistical baselines through reconstruction error threshold tuning.
 - Deployed a **containerized inference service** on GCP using Docker and CI/CD, supporting reliable real-time predictions and safe iterative updates.